

2. **Economic Opportunity:** Agricultural waste has significant economic value when processed correctly. Rice husk, wheat stubble, and sugarcane bagasse can be converted into biofuels, compost, and industrial raw materials.
3. **Technology Gap:** While India has made strides in digital agriculture, there is a notable absence of platforms specifically addressing waste logistics, tracking, and recycling coordination.
4. **Policy Support:** The Indian government's push for sustainable agriculture and programs like the National Policy for Management of Crop Residues provide a supportive policy environment for technology-driven solutions.
5. **Carbon Credit Market:** The growing carbon credit ecosystem offers financial incentives for verifiable waste diversion, which AgriTrace can quantify and track.

1.3 Technology Overview

AgriTrace is built using a modern, production-grade technology stack carefully selected to deliver reliability, scalability, and excellent developer experience. The following is a brief overview of the core technologies used:

- **Next.js 15.5:** A React-based framework providing server-side rendering (SSR), static site generation (SSG), API routes, and file-based routing. Next.js enables building full-stack web applications within a single codebase, eliminating the need for a separate backend server. It supports middleware, incremental static regeneration, and optimized image handling out of the box.
- **TypeScript 5.0:** A statically typed superset of JavaScript that provides compile-time type checking, enhanced IDE support, and improved code maintainability. TypeScript's type system catches errors during development rather than at runtime, significantly reducing bugs in production.
- **Firebase:** Google's Backend-as-a-Service (BaaS) platform, providing:
 - *Cloud Firestore* – A NoSQL document database with real-time synchronization capabilities and offline support.
 - *Firebase Authentication* – Managed identity service supporting email/password, OAuth providers, and email verification.
 - *Firebase Storage* – Cloud-based file storage for waste report photos and delivery proofs.
- **Tailwind CSS:** A utility-first CSS framework that enables rapid UI development through composable utility classes. Combined with a custom design system, Tailwind CSS ensures visual consistency across all components.